

value_name, x
ident
integer_literal
index, i, j, n, m

$typeconstr$	$::=$ unit bool int	
$typvar, tv$	$::=$ $'ident$	
$typexpr, t$	$::=$ $typvar$ $typexpr \rightarrow typexpr'$ $typeconstr$ $(typexpr)$	S
$typscheme, ts$	$::=$ $(typvar_1, \dots, typvar_n)typexpr$ generalise (Γ, t)	bind $typvar_1..typvar_n$ in $typexpr$ M
$constant, c$	$::=$ $integer_literal$ false true $()$ $(\&\&)$ not	
$expr, e$	$::=$ $value_name$ $constant$ $expr\ expr'$ function $value_name \rightarrow expr$ let $value_name = expr$ in $expr'$ $(expr)$ $\{v/x\}e$	bind $value_name$ in $expr$ bind $value_name$ in $expr'$ S M
$value, v$	$::=$ $constant$ function $value_name \rightarrow expr$	
Γ	$::=$ empty $\Gamma, value_name : typscheme$	
$formula$	$::=$ $judgement$ not $(formula)$ $typscheme > t$ $typscheme = typscheme'$ $value_name = value_name'$	

<i>terminals</i>	$::=$ $\rightarrow$ function $\vdash$ $\longrightarrow$ $\{$ $\}$
<i>Jtype</i>	$::=$ $value_name : \textit{typscheme} \textbf{in } \Gamma$ $\Gamma \vdash constant : t$ $\Gamma \vdash e : t$
<i>Jop</i>	$::=$ $e \longrightarrow e'$
<i>judgement</i>	$::=$ $Jtype$ Jop
<i>user_syntax</i>	$::=$ $value_name$ $ident$ $integer_literal$ $index$ $typeconstr$ $typvar$ $typexpr$ $typscheme$ $constant$ $expr$ $value$ Γ $formula$ $terminals$

$value_name : \textit{typscheme} \textbf{in } \Gamma$

$$\frac{}{value_name : \textit{typscheme} \textbf{in } \Gamma, value_name : \textit{typscheme}} \quad \text{VTSIN_VN1}$$

$$\frac{\frac{value_name : \textit{typscheme} \textbf{in } \Gamma \quad \textbf{not } (value_name = value_name')}{value_name : \textit{typscheme} \textbf{in } \Gamma, value_name' : \textit{typscheme}'}}{} \quad \text{VTSIN_VN2}$$

$\Gamma \vdash constant : t$

$$\frac{}{\Gamma \vdash integer_literal : \textbf{int}} \quad \text{CONSTANT_INT}$$

$$\frac{}{\Gamma \vdash \textbf{false} : \textbf{bool}} \quad \text{CONSTANT_FALSE}$$

$$\frac{}{\Gamma \vdash \textbf{true} : \textbf{bool}} \quad \text{CONSTANT_TRUE}$$

$$\frac{}{\Gamma \vdash () : \mathbf{unit}} \quad \text{CONSTANT_UNIT}$$

$$\frac{}{\Gamma \vdash (\&\&) : \mathbf{bool} \rightarrow (\mathbf{bool} \rightarrow \mathbf{bool})} \quad \text{CONSTANT_AND}$$

$$\frac{}{\Gamma \vdash \mathbf{not} : \mathbf{bool} \rightarrow \mathbf{bool}} \quad \text{CONSTANT_NOT}$$

$$\boxed{\Gamma \vdash e : t}$$

$$\frac{x : \textit{typscheme} \textbf{in} \Gamma \quad \textit{typscheme} > t}{\Gamma \vdash x : t} \quad \text{GET_VALUE_NAME}$$

$$\frac{\Gamma \vdash \textit{constant} : t}{\Gamma \vdash \textit{constant} : t} \quad \text{GET_CONSTANT}$$

$$\frac{\Gamma \vdash e : t_1 \rightarrow t_2 \quad \Gamma \vdash e' : t_1}{\Gamma \vdash e e' : t_2} \quad \text{GET_APPLY}$$

$$\frac{\Gamma, x_1 : () t_1 \vdash e : t}{\Gamma \vdash \mathbf{function} x_1 \rightarrow e : t_1 \rightarrow t} \quad \text{GET_LAMBDA}$$

$$\frac{\Gamma \vdash e : t \quad \Gamma, x : \textit{typscheme} \vdash e' : t' \quad \textit{typscheme} = \mathbf{generalise}(\Gamma, t)}{\Gamma \vdash \mathbf{let} x = e \textbf{in} e' : t'} \quad \text{GET_LET}$$

$$\boxed{e \longrightarrow e'}$$

$$\frac{}{(\mathbf{function} x \rightarrow e) v \longrightarrow \{v/x\}e} \quad \text{JO_RED_APP}$$

$$\frac{}{\mathbf{let} x = v \textbf{in} e \longrightarrow \{v/x\}e} \quad \text{JO_RED_LET}$$

$$\frac{e \longrightarrow e'}{e e_1 \longrightarrow e' e_1} \quad \text{JO_RED_CONTEXT_APP1}$$

$$\frac{e \longrightarrow e'}{v e \longrightarrow v e'} \quad \text{JO_RED_CONTEXT_APP2}$$

$$\frac{e \longrightarrow e'}{\mathbf{let} x = e \textbf{in} e_1 \longrightarrow \mathbf{let} x = e' \textbf{in} e_1} \quad \text{JO_RED_CONTEXT_LET}$$

$$\frac{}{\mathbf{not} \mathbf{true} \longrightarrow \mathbf{false}} \quad \text{JO_RED_NOT_1}$$

$$\frac{}{\mathbf{not} \mathbf{false} \longrightarrow \mathbf{true}} \quad \text{JO_RED_NOT_2}$$

$$\frac{}{((\&\&) \mathbf{true}) e \longrightarrow e} \quad \text{JO_RED_AND_1}$$

$$\frac{}{((\&\&) \mathbf{false}) e \longrightarrow \mathbf{false}} \quad \text{JO_RED_AND_2}$$

Definition rules: 22 good 0 bad

Definition rule clauses: 36 good 0 bad